ZentroMart VoiceOps agent: report

What I built

Step 1. One LLM call with a system prompt (role, limits, AI-assistant introduction). No documents.
Step 2. RAG over kb/*.md (one chunk per [SECTION-ID], status and effective_from kept) and the product catalog; BM25 with a Hinglish-to-English query bridge, SKU matching, catalog-to-policy joins and KB-00.1 precedence weighting. Answers use only retrieved excerpts, say "I don't have that information" otherwise, and log the chunk IDs used.
Step 3. A tool-calling conversation agent: history, order/ticket tools, code-enforced identity check with one retry, code-computed business-day and return-window dates, escalation flags, and checks that a handover is only claimed after create_ticket returned created=True.
Step 4. Gnani STT (ITN on, Hindi fallback) and TTS (timbre-v2.5), spoken-digit normalisation, ear-friendly reply text, batch runner (run.py) and a browser demo (app.py).

Step 1 vs Step 2 (practice questions, 52)

run	correct /52	accuracy	errors (rate limit)	facts included	decision ok	evidence recall	"I don't know" cases
Step 1: prompt only	5	0.096	3	0.143	0.327	0.102	0/5
Step 2: RAG	38	0.731	0	0.769	0.942	0.971	5/5
Step 3 agent, single questions	38	0.731	3	0.837	0.837	0.98	4/5
Step 2 on ASR-style text	*not run*						

Accuracy = must_include satisfied, no must_not_include phrase, acceptable decision.

Retrieval only (no LLM, text queries, k=6)

questions with gold evidence: 47 (k=6, text queries)
mean recall@k : 0.968
all-gold-found rate: 0.936
MRR (first gold) : 0.798
hard-negative hit : 0.064

Step 3 conversations (23 conversations, 96 turns)

run	turn accuracy	conversations fully correct	decision ok	tool calls ok	facts included
Step 3 agent	*not run*				

What went wrong

Evaluation limits (please read). I used Groq's free tier (gpt-oss-120b, 200,000 tokens/day). When it ran out, calls rotated to gpt-oss-20b and qwen3.8-27b, and --resume merged runs made on different days, so the Step 2 and Step 3 rows mix models and two prompt versions. The Step 1 row used gpt-oss-120b only. 3 Step 1 and 3 full-agent questions errored (rate limit / malformed tool call) and count as wrong. The quota ran out before I could run the 23-conversation evaluation, so Step 3 conversation handling is **not scored**, and my end-to-end voice batch got only quota-error apologies from the LLM (Gnani STT/TTS themselves worked: my clips were transcribed and replies were synthesised).
Step 1. Without documents the model guesses: 14% of required facts present, and 0 of 5 questions the documents do not cover were answered with 'I don't know'. Step 2 got 5 of 5.
Step 2 failures (14). 6 wrong refund deadlines: the LLM miscounts business days (Sundays and the 19 October holiday). 3 warranty answers omitted the 12-month period. Hinglish questions answered in Devanagari. 'Clarify' chosen after already answering. Swimsuit/briefs not stated as 'cannot be returned'.
Full agent on single questions (14 wrong, 3 errors). Date maths fixed by the refund_eta tool, but new problems: wrong decision label on escalation/safety questions (PQ0023, PQ0030, PQ0031), and 2 questions failed because the model sent null for an optional tool argument.

What I would fix

Fixed in code after those runs but **not re-scored** (quota): null tool arguments are tolerated; the 'I am an AI assistant' opening is enforced in code; empty replies are retried; retry-after is honoured. Next (in order): re-run all evaluations on one model with a larger quota; tighten escalation prompts for safety/unauthorised-activity cases; add multilingual embeddings for Hindi; shorten prompts to cut tokens per call.

Bonus work

Hybrid BM25 + optional multilingual dense retrieval; query bridge for Hinglish; catalog-to-policy joins; code-enforced identity check with one retry and ticket honesty; spoken-digit normalisation for ASR text; offline tests with a scripted LLM (9 passing); resumable evaluation runner; browser demo with microphone recording.